

SH-Waves: Reflection and Transmission Coefficients, Energy Conservation and Partitioning at an Interface, and Post-Critical Reflection

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This tutorial is in **3 parts**. The **first part** presents the derivation for the Reflection and Transmission coefficients for SH waves, the **second part** discusses the energy conservation and partitioning for SH-waves at an interface, and the **final part** presents the mathematical analysis of post-critical SH-wave reflection.

Part 1 – SH-Wave Reflection and Transmission Coefficients

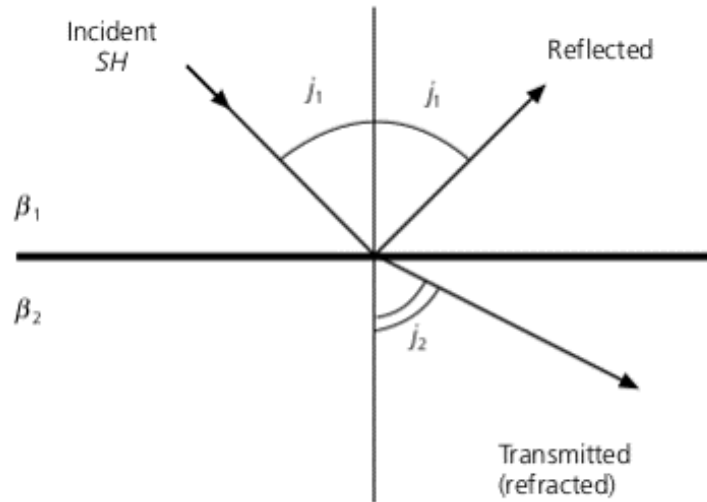


Figure 1: SH R/T Coefficients

Problem Setup and Wave Equations

We have two **homogeneous, isotropic elastic solids** in perfect contact at the plane $z = 0$.

- **Medium 1** ($z < 0$): Density ρ_1 , Shear Modulus μ_1 , S-wave velocity $\beta_1 = \sqrt{\frac{\mu_1}{\rho_1}}$
- **Medium 2** ($z > 0$): Density ρ_2 , Shear Modulus μ_2 , S-wave velocity $\beta_2 = \sqrt{\frac{\mu_2}{\rho_2}}$

An **incident SH-plane wave** propagates in Medium 1 towards the interface with angle j_1 from the normal. This generates:

1. A **reflected SH-wave** in Medium 1 at angle j_1
2. A **transmitted SH-wave** in Medium 2 at angle j_2 , governed by Snell's Law:

$$\frac{\sin j_1}{\beta_1} = \frac{\sin j_2}{\beta_2} = p$$

where p is the ray parameter.

The displacement for SH-waves is purely in the y -direction, u_y . So, the particle motion is horizontal (in the y -direction), perpendicular to both the propagation direction and the interface normal.

The general form of a plane wave is:

$$u_y = A e^{i(\omega t - \mathbf{k} \cdot \mathbf{r})} = A e^{i(\omega t - k_x x - k_z z)}$$

Let's define the waves explicitly:

- **Incident Wave** (downgoing in Medium 1, amplitude A_1):

$$u_y^{\text{inc}} = A_1 e^{i(\omega t - k_x x - k_{z\beta_1} z)}$$

- **Reflected Wave** (upgoing in Medium 1, amplitude A'):

$$u_y^{\text{ref}} = A' e^{i(\omega t - k_x x + k_{z\beta_1} z)}$$

(Note the + sign in front of $k_{z\beta_1} z$, indicating propagation in the $+z$ direction)

- **Transmitted Wave** (downgoing in Medium 2, amplitude A_2):

$$u_y^{\text{trans}} = A_2 e^{i(\omega t - k_x x - k_{z\beta_2} z)}$$

The **total wavefield** in each medium is:

- **In Medium 1** ($z < 0$): $u_{y1} = u_y^{\text{inc}} + u_y^{\text{ref}}$

$$u_{y1} = A_1 e^{i(\omega t - k_x x - k_{z\beta_1} z)} + A' e^{i(\omega t - k_x x + k_{z\beta_1} z)}$$

- **In Medium 2** ($z > 0$): $u_{y2} = u_y^{\text{trans}}$

$$u_{y2} = A_2 e^{i(\omega t - k_x x - k_{z\beta_2} z)}$$

From the wave equation and geometry, the **wavenumbers** are:

$$\begin{aligned} |\mathbf{k}_1| &= \frac{\omega}{\beta_1}, & |\mathbf{k}_2| &= \frac{\omega}{\beta_2} \\ k_x &= |\mathbf{k}_1| \sin j_1 = \frac{\omega}{\beta_1} \sin j_1 \\ k_{z\beta_1} &= |\mathbf{k}_1| \cos j_1 = \frac{\omega}{\beta_1} \cos j_1 \\ k_{z\beta_2} &= |\mathbf{k}_2| \cos j_2 = \frac{\omega}{\beta_2} \cos j_2 \end{aligned}$$

where:

- A_1 = incident wave amplitude
- A' = reflected wave amplitude
- A_2 = transmitted wave amplitude
- k_x = horizontal wavenumber (same in both media by Snell's law i.e $k_{z\alpha_1} = k_{z\alpha_2}$)
- $k_{z\beta_i}$ = vertical wavenumber in medium i
- j = the angle from the vertical (normal to interface)

Boundary Conditions

At the interface $z = 0$, the physics requires continuity of two fields:

1. **Continuity of Displacement** (u_y): The particles on both sides of the boundary must move together.

$$u_{y1}(z = 0) = u_{y2}(z = 0)$$

2. **Continuity of Stress** (σ_{yz}): The shear traction acting on the interface must be continuous.

Remember for an isotropic homogeneous medium, the stress tensor is:

$$\sigma_{ij} = \lambda \delta_{ij} \theta + 2\mu e_{ij}$$

The stress component relevant for an SH-wave is σ_{yz} . For a linear elastic solid:

$$\sigma_{yz} = 2\mu e_{yz}$$

where e_{yz} is the shear strain component, $e_{yz} = \frac{1}{2}(\partial_y u_z + \partial_z u_y)$. Since $u_z = 0$ for SH-waves, this simplifies to:

$$\sigma_{yz} = 2\mu e_{yz} = 2\mu \frac{1}{2}(\partial_y u_z + \partial_z u_y) = \mu \left(\frac{\partial u_z}{\partial y} + \frac{\partial u_y}{\partial z} \right)$$

$$\boxed{\sigma_{yz} = \mu \frac{\partial u_y}{\partial z}}$$

Thus, the boundary condition is:

$$\sigma_{yz}^{(1)}(z=0) = \sigma_{yz}^{(2)}(z=0) \quad \Rightarrow \quad \mu_1 \frac{\partial u_{y1}}{\partial z} \Big|_{z=0} = \mu_2 \frac{\partial u_{y2}}{\partial z} \Big|_{z=0}$$

Applying the Boundary Conditions

Condition 1: Continuity of Displacement at $z = 0$

$$u_{y1}(z=0) = u_{y2}(z=0)$$

$$A_1 e^{i(\omega t - k_x x - k_{z\beta_1} \cdot 0)} + A' e^{i(\omega t - k_x x + k_{z\beta_1} \cdot 0)} = A_2 e^{i(\omega t - k_x x - k_{z\beta_2} \cdot 0)}$$

$$[A_1 e^{i(\omega t - k_x x)} + A' e^{i(\omega t - k_x x)}] = [A_2 e^{i(\omega t - k_x x)}]$$

The exponential terms $e^{i(\omega t - k_x x)}$ are identical on all terms (**this is the phase matching condition from Snell's Law**) and can be canceled out. This leaves us with a simple equation for the amplitudes:

$$\boxed{A_1 + A' = A_2} \quad (1)$$

Condition 2: Continuity of Stress at $z = 0$

First, compute the stress in each medium by taking the spatial derivative ∂_z .

Stress in Medium 1 ($\sigma_{yz}^{(1)}$)

$$\begin{aligned}
u_{y1} &= A_1 e^{i(\omega t - k_x x - k_{z\beta_1} z)} + A' e^{i(\omega t - k_x x + k_{z\beta_1} z)} \\
\frac{\partial u_{y1}}{\partial z} &= -ik_{z\beta_1} A_1 e^{i(\omega t - k_x x - k_{z\beta_1} z)} + ik_{z\beta_1} A' e^{i(\omega t - k_x x + k_{z\beta_1} z)} \\
\left. \frac{\partial u_{y1}}{\partial z} \right|_{z=0} &= -ik_{z\beta_1} A_1 e^{i(\omega t - k_x x)} + ik_{z\beta_1} A' e^{i(\omega t - k_x x)} \\
\left. \sigma_{yz}^{(1)} \right|_{z=0} &= \mu_1 \left. \frac{\partial u_{y1}}{\partial z} \right|_{z=0} = -i\mu_1 k_{z\beta_1} A_1 e^{i(\omega t - k_x x)} + i\mu_1 k_{z\beta_1} A' e^{i(\omega t - k_x x)}
\end{aligned}$$

Stress in Medium 2 ($\sigma_{yz}^{(2)}$)

$$\begin{aligned}
u_{y2} &= A_2 e^{i(\omega t - k_x x - k_{z\beta_2} z)} \\
\frac{\partial u_{y2}}{\partial z} &= -ik_{z\beta_2} A_2 e^{i(\omega t - k_x x - k_{z\beta_2} z)} \\
\left. \frac{\partial u_{y2}}{\partial z} \right|_{z=0} &= -ik_{z\beta_2} A_2 e^{i(\omega t - k_x x)} \\
\left. \sigma_{yz}^{(2)} \right|_{z=0} &= \mu_2 \left. \frac{\partial u_{y2}}{\partial z} \right|_{z=0} = -i\mu_2 k_{z\beta_2} A_2 e^{i(\omega t - k_x x)}
\end{aligned}$$

Now apply the boundary condition $\sigma_{yz}^{(1)} = \sigma_{yz}^{(2)}$ at $z = 0$:

$$-i\mu_1 k_{z\beta_1} A_1 e^{i(\omega t - k_x x)} + i\mu_1 k_{z\beta_1} A' e^{i(\omega t - k_x x)} = -i\mu_2 k_{z\beta_2} A_2 e^{i(\omega t - k_x x)}$$

We can cancel the common factor $-ie^{i(\omega t - k_x x)}$ from all terms:

$$\mu_1 k_{z\beta_1} A_1 - \mu_1 k_{z\beta_1} A' = \mu_2 k_{z\beta_2} A_2$$

Dividing through by $\mu_1 k_{z\beta_1}$:

$$A_1 - A' = \frac{\mu_2 k_{z\beta_2}}{\mu_1 k_{z\beta_1}} A_2$$

$$\boxed{A_1 - A' = \frac{\mu_2 k_{z\beta_2}}{\mu_1 k_{z\beta_1}} A_2} \quad (2)$$

Solving for the R/T Coefficients

We now have a system of two equations with two unknowns (A' and A_2), with A_1 as the known incident amplitude.

From Eq. (1): $A_2 = A_1 + A'$

Substitute this into Eq. (2):

$$A_1 - A' = \frac{\mu_2 k_{z\beta_2}}{\mu_1 k_{z\beta_1}} (A_1 + A')$$

Let $M = \frac{\mu_2 k_{z\beta_2}}{\mu_1 k_{z\beta_1}}$ for simplicity.

$$A_1 - A' = M(A_1 + A')$$

$$A_1 - A' = MA_1 + MA'$$

$$A_1 - MA_1 = A' + MA'$$

$$A_1(1 - M) = A'(1 + M)$$

Reflection Coefficient ($R_{12} = \frac{A'}{A_1}$)

$$R_{12} = \frac{A'}{A_1} = \frac{1 - M}{1 + M}$$

Substituting back for M :

$$R_{12} = \frac{1 - \frac{\mu_2 k_{z\beta_2}}{\mu_1 k_{z\beta_1}}}{1 + \frac{\mu_2 k_{z\beta_2}}{\mu_1 k_{z\beta_1}}} = \frac{\mu_1 k_{z\beta_1} - \mu_2 k_{z\beta_2}}{\mu_1 k_{z\beta_1} + \mu_2 k_{z\beta_2}} \quad (3)$$

This is the first result. We can express it in terms of impedance $\rho\beta$ and angles. Recall:

$$\mu = \rho\beta^2 \quad \text{and} \quad k_{z\beta} = \frac{\omega}{\beta} \cos j$$

Substitute into Eq. (3):

$$R_{12} = \frac{(\rho_1 \beta_1^2) \left(\frac{\omega}{\beta_1} \cos j_1 \right) - (\rho_2 \beta_2^2) \left(\frac{\omega}{\beta_2} \cos j_2 \right)}{(\rho_1 \beta_1^2) \left(\frac{\omega}{\beta_1} \cos j_1 \right) + (\rho_2 \beta_2^2) \left(\frac{\omega}{\beta_2} \cos j_2 \right)}$$

Simplify by canceling β and ω :

$$R_{12} = \frac{\rho_1 \beta_1 \cos j_1 - \rho_2 \beta_2 \cos j_2}{\rho_1 \beta_1 \cos j_1 + \rho_2 \beta_2 \cos j_2}$$

Transmission Coefficient ($T_{12} = \frac{A_2}{A_1}$)

From Eq. (1):

$$T_{12} = \frac{A_2}{A_1} = \frac{A_1 + A'}{A_1} = 1 + \frac{A'}{A_1} = 1 + R_{12}$$

$$T_{12} = 1 + \frac{\mu_1 k_{z\beta_1} - \mu_2 k_{z\beta_2}}{\mu_1 k_{z\beta_1} + \mu_2 k_{z\beta_2}} = \frac{(\mu_1 k_{z\beta_1} + \mu_2 k_{z\beta_2}) + (\mu_1 k_{z\beta_1} - \mu_2 k_{z\beta_2})}{\mu_1 k_{z\beta_1} + \mu_2 k_{z\beta_2}}$$

$$T_{12} = \frac{2\mu_1 k_{z\beta_1}}{\mu_1 k_{z\beta_1} + \mu_2 k_{z\beta_2}} \quad (4)$$

Again, we may express it in terms of impedance and angles:

$$T_{12} = \frac{2(\rho_1 \beta_1^2) \left(\frac{\omega}{\beta_1} \cos j_1 \right)}{(\rho_1 \beta_1^2) \left(\frac{\omega}{\beta_1} \cos j_1 \right) + (\rho_2 \beta_2^2) \left(\frac{\omega}{\beta_2} \cos j_2 \right)}$$

Simplify:

$$T_{12} = \frac{2\rho_1 \beta_1 \cos j_1}{\rho_1 \beta_1 \cos j_1 + \rho_2 \beta_2 \cos j_2}$$

Key Properties:

- $T_{12} = 1 + R_{12}$ (amplitude conservation in form)
- At normal incidence ($j_1 = j_2 = 0$): $R_{12} = \frac{Z_1 - Z_2}{Z_1 + Z_2}$ where $Z = \rho\beta$
- Energy conservation requires consideration of impedances and angles (not just amplitudes)

Summary and Physical Interpretation

- **Reflection Coefficient (R_{12}):** Governs the amplitude of the reflected wave relative to the incident wave. It depends on the contrast in the shear impedance ($\rho\beta$) projected onto the vertical direction ($\rho\beta \cos j$), often called the vertical impedance.
 - If $\rho_1 \beta_1 \cos j_1 = \rho_2 \beta_2 \cos j_2$, there is no reflection ($R_{12} = 0$), even if the media are different. This is known as a impedance match.
 - If the wave goes into a "harder" medium (higher vertical impedance), R_{12} is negative, meaning the reflection has a phase flip.
- **Transmission Coefficient (T_{12}):** Governs the amplitude of the transmitted wave. It is always positive, indicating no phase change upon transmission for SH-waves.

These coefficients satisfy energy conservation. The relative energy flux (related to $\rho\beta \times \text{amplitude}^2$) of the reflected and transmitted waves sums to the energy of the incident wave.

Part 2 – Energy Flux and Partition for SH Waves

1. Mechanical Energy Density for Plane Waves

For elastic waves, the mechanical energy density consists of **kinetic** and **potential (strain)** energy:

$$E = E_{\text{kinetic}} + E_{\text{potential}} \quad (1)$$

Kinetic Energy Density

For a plane wave with particle velocity $\dot{u} = \frac{\partial u}{\partial t}$:

$$u = Ae^{i(\omega t - \mathbf{k} \cdot \mathbf{x})} \quad (2)$$

The particle velocity is:

$$\dot{u} = \frac{\partial u}{\partial t} = i\omega Ae^{i(\omega t - \mathbf{k} \cdot \mathbf{x})} \quad (3)$$

Kinetic energy density (per unit volume):

$$KE = \frac{1}{2}\rho\dot{u}^2 \quad (4)$$

$$= \frac{1}{2}\rho \left(\frac{\partial \mathbf{u}}{\partial t} \right)^2 \quad (5)$$

Taking the time average over one period (noting that $\langle e^{i\omega t} e^{-i\omega t} \rangle = 1$ and $\langle e^{2i\omega t} \rangle = 0$):

For complex notation, the time-averaged kinetic energy density is:

$$\langle E_{\text{kinetic}} \rangle = \frac{1}{2}\rho \langle (i\omega A)^2 \rangle = \frac{1}{2}\rho\omega^2 A^2 \cdot \frac{1}{2} = \frac{\rho\omega^2 A^2}{4} \quad (6)$$

where the factor of 1/2 comes from time averaging $\cos^2$ or $\sin^2$ over a period.

$$\langle KE \rangle = \frac{1}{2}\rho \frac{|\dot{u}|^2}{2} = \frac{1}{4}\rho\omega^2 |A|^2 \quad (7)$$

Potential (Strain) Energy Density

General Formula for Strain Energy Density:

The **strain energy density** (elastic potential energy density) is defined as the work done per unit volume to deform the body from zero strain to its current strain state:

$$w = \int_0^\varepsilon \boldsymbol{\sigma} : d\boldsymbol{\varepsilon}.$$

For a linear elastic material (Hooke's law), this reduces to

$$w = \frac{1}{2} \boldsymbol{\sigma} : \boldsymbol{\varepsilon},$$

where

- w = strain energy density (J/m³),
- $\boldsymbol{\sigma}$ = stress tensor,
- $\boldsymbol{\varepsilon}$ = strain tensor,
- “:” denotes the double contraction, i.e. $\boldsymbol{\sigma} : \boldsymbol{\varepsilon} = \sum_{i,j} \sigma_{ij} \varepsilon_{ij}$.

For shear waves, the strain energy density is:

$$E_{\text{potential}} = \frac{1}{2} \mu \epsilon_{ij} \epsilon_{ij} \quad (8)$$

For SH waves with displacement $u_y = A e^{i(\omega t - k_x x - k_z z)}$:

$$\epsilon_{yz} = \frac{1}{2} \left(\frac{\partial u_y}{\partial z} + \frac{\partial u_z}{\partial y} \right) = \frac{1}{2} \frac{\partial u_y}{\partial z} \quad (9)$$

$$\frac{\partial u_y}{\partial z} = -i k_z A e^{i(\omega t - k_x x - k_z z)} \quad (10)$$

Potential Energy Density

The instantaneous potential energy density for shear waves is:

$$E_{\text{potential}} = \frac{1}{2} \mu (\partial_z u_y)^2 \quad (11)$$

The time-averaged potential energy density is:

$$\langle E_{\text{potential}} \rangle = \frac{1}{2} \mu \langle (\partial_z u_y)^2 \rangle \quad (12)$$

To compute the time average, we first consider the real part of the displacement field:

$$u_y^{\text{real}} = \text{Re}[A e^{i(\omega t - k_x x - k_z z)}] = A \cos(\omega t - k_x x - k_z z) \quad (13)$$

$$\frac{\partial u_y^{\text{real}}}{\partial z} = A k_z \sin(\omega t - k_x x - k_z z) \quad (14)$$

$$\left(\frac{\partial u_y^{\text{real}}}{\partial z} \right)^2 = A^2 k_z^2 \sin^2(\omega t - k_x x - k_z z) \quad (15)$$

Time Averaging of Trigonometric function

The time average of $\sin^2$ over one period $T = \frac{2\pi}{\omega}$ is:

$$\langle \sin^2(\omega t + \phi) \rangle = \frac{1}{T} \int_0^T \sin^2(\omega t + \phi) dt = \frac{1}{2} \quad (16)$$

where $\phi = -k_x x - k_z z$ is the spatial phase.

The time-averaged potential energy density (considering the strain tensor properly):

$$\langle E_{\text{potential}} \rangle = \frac{1}{2} \mu \langle (\partial_z u_y)^2 \rangle = \frac{1}{2} \mu \cdot A^2 k_z^2 \cdot \frac{1}{2} \quad (17)$$

$$= \frac{\mu k_z^2 A^2}{4} \quad (18)$$

Take Note:

The phase disappears because:

- Time-averaging over a full period makes any oscillatory term with $e^{\pm 2i\omega t}$ average to zero
- The $\frac{1}{2}$ factor comes from $\langle \cos^2 \rangle = \langle \sin^2 \rangle = \frac{1}{2}$
- The spatial phase $-k_x x - k_z z$ becomes a constant during time-averaging at a fixed position

Total Mechanical Energy Density

Using the dispersion relation for shear waves: $k^2 = k_x^2 + k_z^2 = \omega^2/\beta^2$ and $\mu = \rho\beta^2$:

$$E = \langle KE \rangle + \langle PE \rangle \quad (19)$$

$$\langle E_{\text{total}} \rangle = \frac{\rho\omega^2 A^2}{4} + \frac{\mu k^2 A^2}{4} = \frac{\rho\omega^2 A^2}{4} + \frac{\rho\beta^2 \cdot \omega^2/\beta^2 \cdot A^2}{4} \quad (20)$$

$$\langle E_{\text{total}} \rangle = \frac{\rho\omega^2 A^2}{4} + \frac{\rho\omega^2 A^2}{4} = \frac{\rho\omega^2 A^2}{2} \quad (21)$$

Result (per unit wavelength λ):

$$\boxed{E = \frac{\rho\omega^2 A^2}{2}} \quad (22)$$

2. Energy Flux Along Ray Direction

Energy flux is the rate of energy flow per unit area, also called intensity or energy current density.

Energy Flux Density (Intensity)

The energy flux, $\mathbf{J}$, (power per unit area) is given by the stress-velocity product. For a wave propagating with group velocity $\mathbf{v}_g$:

$$\mathbf{J} = -\langle \sigma_{ij} \dot{u}_j \rangle \quad (23)$$

$$\mathbf{J} = -\boldsymbol{\sigma} \cdot \dot{\mathbf{u}} \quad (24)$$

For SH waves propagating in direction making angle j with the vertical (z-axis):

$$J = \sigma_{yz} \cdot \frac{\partial u_y}{\partial t} \quad (25)$$

Computing the Energy Flux

For the wave $u_y = Ae^{i(\omega t - k_x x - k_z z)}$:

$$\frac{\partial u_y}{\partial t} = i\omega Ae^{i(\omega t - k_x x - k_z z)} \quad (26)$$

$$\frac{\partial u_y}{\partial z} = -ik_z A \exp i(\omega t - k_x x - k_z z) \quad (27)$$

$$\sigma_{yz} = \mu \frac{\partial u_y}{\partial z} = -i\mu k_z A e^{i(\omega t - k_x x - k_z z)} \quad (28)$$

The instantaneous energy flux is:

$$J = \sigma_{yz} \cdot \frac{\partial u_y}{\partial t} = (-i\mu k_z A)(i\omega A)e^{i(\omega t - k_x x - k_z z)} \quad (29)$$

$$J = \mu\omega k_z A^2 e^{2i(\omega t - k_x x - k_z z)} \quad (30)$$

Time-averaging over one period:

$$\langle J \rangle = \mu\omega k_z A^2 \cdot \frac{1}{2} = \frac{\mu\omega k_z A^2}{2} \quad (31)$$

Using $\mu = \rho\beta^2$ and $k_z = k \cos j = \frac{\omega}{\beta} \cos j$:

$$\langle J \rangle = \frac{\rho\beta^2\omega \cdot \frac{\omega}{\beta} \cos j \cdot A^2}{2} = \frac{\rho\omega^2\beta A^2 \cos j}{2} \quad (32)$$

Since the energy flux is in the direction of the ray (velocity $v = \beta$):

$$J = \frac{\rho\omega^2 A^2}{2} v \cos j \cdot dS = \frac{\rho\omega^2 A^2}{2} \beta \cos j \cdot dS \quad (33)$$

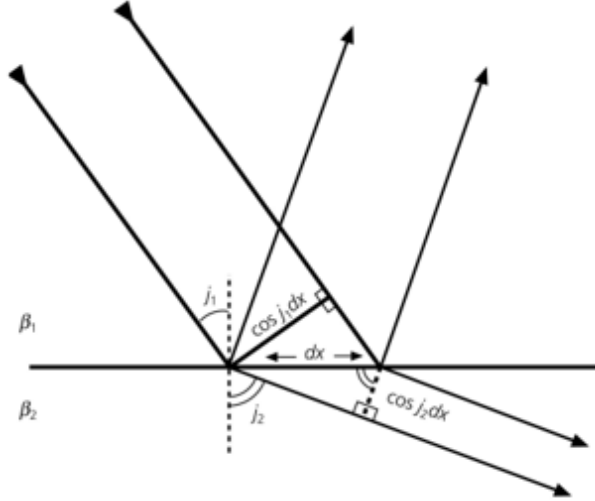
Or more generally along the ray:

$$J = \frac{\rho\omega^2 A^2}{2} v dS \quad (34)$$

This represents energy flow through a surface perpendicular to the propagation direction. Where dS is the area element perpendicular to the ray.

3. Energy Flux Across the Interface

Consider energy flux passing through the horizontal interface (perpendicular to the z -axis). The interface is horizontal, but the wave propagates at an angle j to the vertical.



Geometric Consideration

For a ray traveling at angle j from the vertical (z -axis), the vertical component of velocity is $v \cos j$.

The energy flux crossing a horizontal area element dx (per unit length in the y -direction) is:

$$J = \frac{\rho\omega^2 A^2}{2} v \cos j dx \quad (35)$$

This represents the energy passing through the interface per unit time.

4. Energy Flux for Incident, Reflected, and Transmitted Waves

For SH waves at an interface:

Incident Wave

Wave amplitude: A_1 , velocity: β_1 , angle: j_1 , density: ρ_1

$$\boxed{J_I = \frac{\omega^2 A_1^2 \rho_1 \beta_1 \cos j_1 dx}{2}} \quad (36)$$

Reflected Wave

The reflected wave has amplitude $A' = A_1 R_{12}$, travels with velocity β_1 , at angle j_1 (but upward):

$$J_R = \frac{\omega^2 (A')^2 \rho_1 \beta_1 \cos j_1 dx}{2} = \frac{\omega^2 A_1^2 R_{12}^2 \rho_1 \beta_1 \cos j_1 dx}{2} \quad (37)$$

$$\boxed{J_R = \frac{\omega^2 A_1^2 R_{12}^2 \rho_1 \beta_1 \cos j_1 dx}{2}} \quad (38)$$

Transmitted Wave

The transmitted wave has amplitude $A_2 = A_1 T_{12}$, travels with velocity β_2 , at angle j_2 , density: ρ_2

$$J_T = \frac{\omega^2 A_2^2 \rho_2 \beta_2 \cos j_2 dx}{2} = \frac{\omega^2 A_1^2 T_{12}^2 \rho_2 \beta_2 \cos j_2 dx}{2} \quad (39)$$

$$\boxed{J_T = \frac{\omega^2 A_1^2 T_{12}^2 \rho_2 \beta_2 \cos j_2 dx}{2}} \quad (40)$$

5. Energy Conservation

Energy conservation requires that the incident energy equals the sum of reflected and transmitted energy. So, energy conservation at the interface:

$$J_I = J_R + J_T \quad (41)$$

This states that the incident energy flux equals the sum of reflected and transmitted energy fluxes.

Substituting the expressions:

$$\frac{\rho_1 \omega^2 A_1^2}{2} \beta_1 \cos j_1 dx = \frac{\rho_1 \omega^2 A_1^2 R_{12}^2}{2} \beta_1 \cos j_1 dx + \frac{\rho_2 \omega^2 A_1^2 T_{12}^2}{2} \beta_2 \cos j_2 dx$$

Canceling common factors $\frac{\omega^2 A_1^2}{2} dx$:

$$\rho_1 \beta_1 \cos j_1 = \rho_1 \beta_1 \cos j_1 R_{12}^2 + \rho_2 \beta_2 \cos j_2 T_{12}^2$$

Dividing by $\rho_1\beta_1 \cos j_1$:

$$1 = R_{12}^2 + T_{12}^2 \frac{\rho_2\beta_2 \cos j_2}{\rho_1\beta_1 \cos j_1} \quad (42)$$

6. Energy Partition Coefficients

Reflected Energy Ratio

The fraction of incident energy that is reflected:

$$\frac{J_R}{J_I} = \frac{\omega^2 A_1^2 R_{12}^2 \rho_1 \beta_1 \cos j_1 \frac{dx}{2}}{\omega^2 A_1^2 \rho_1 \beta_1 \cos j_1 \frac{dx}{2}} \quad (43)$$

Simplifying:

$$\frac{J_R}{J_I} = R_{12}^2 \quad (44)$$

Transmitted Energy Ratio

The fraction of incident energy that is transmitted:

$$\frac{J_T}{J_I} = \frac{\omega^2 A_1^2 T_{12}^2 \rho_2 \beta_2 \cos j_2 \frac{dx}{2}}{\omega^2 A_1^2 \rho_1 \beta_1 \cos j_1 \frac{dx}{2}} \quad (45)$$

Simplifying:

$$\frac{J_T}{J_I} = T_{12}^2 \frac{\rho_2 \beta_2 \cos j_2}{\rho_1 \beta_1 \cos j_1} = T_{12}^2 \frac{I_2 \cos j_2}{I_1 \cos j_1} \quad (46)$$

where $I_i = \rho_i \beta_i$ is the impedance of medium i .

7. Verification of Energy Conservation

For energy conservation: $\frac{J_R}{J_I} + \frac{J_T}{J_I} = 1$

Let's verify this using our expressions for R_{12} and T_{12} .

Step 1: Express reflection coefficient squared

From the previous derivation:

$$R_{12} = \frac{\mu_1 k_{z\beta_1} - \mu_2 k_{z\beta_2}}{\mu_1 k_{z\beta_1} + \mu_2 k_{z\beta_2}} = \frac{I_1 \cos j_1 - I_2 \cos j_2}{I_1 \cos j_1 + I_2 \cos j_2} \quad (47)$$

where $I_i = \rho_i \beta_i$ and $\mu_i k_{z\beta_i} = \rho_i \beta_i \omega \cos j_i$.

Let $Z_1 = I_1 \cos j_1 = \rho_1 \beta_1 \cos j_1$ and $Z_2 = I_2 \cos j_2 = \rho_2 \beta_2 \cos j_2$ for simplicity.

$$R_{12} = \frac{Z_1 - Z_2}{Z_1 + Z_2} \quad (48)$$

$$R_{12}^2 = \left(\frac{Z_1 - Z_2}{Z_1 + Z_2} \right)^2 = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2} \quad (49)$$

Energy Reflection Coefficient:

$$E_R = \frac{J_R}{J_I} = R_{12}^2 = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$$

Step 2: Express transmission coefficient squared

$$T_{12} = \frac{2\mu_1 k_{z\beta_1}}{\mu_1 k_{z\beta_1} + \mu_2 k_{z\beta_2}} = \frac{2Z_1}{Z_1 + Z_2} \quad (50)$$

$$T_{12}^2 = \left(\frac{2Z_1}{Z_1 + Z_2} \right)^2 = \frac{4Z_1^2}{(Z_1 + Z_2)^2} \quad (51)$$

Energy Transmission Coefficient:

$$E_T = T_{12}^2 \frac{I_2 \cos j_2}{I_1 \cos j_1} = \left(\frac{2I_1 \cos j_1}{I_1 \cos j_1 + I_2 \cos j_2} \right)^2 \frac{I_2 \cos j_2}{I_1 \cos j_1} = \frac{4Z_1^2}{(Z_1 + Z_2)^2} \cdot \frac{Z_2}{Z_1}$$

Step 3: Calculate transmitted energy ratio

$$\frac{J_T}{J_I} = T_{12}^2 \frac{Z_2}{Z_1} = \frac{4Z_1^2}{(Z_1 + Z_2)^2} \cdot \frac{Z_2}{Z_1} = \frac{4Z_1 Z_2}{(Z_1 + Z_2)^2} \quad (52)$$

Step 4: Verify energy conservation

$$\frac{J_R}{J_I} + \frac{J_T}{J_I} = R_{12}^2 + T_{12}^2 \frac{Z_2}{Z_1} \quad (53)$$

$$= \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2} + \frac{4Z_1 Z_2}{(Z_1 + Z_2)^2} \quad (54)$$

$$= \frac{(Z_1 - Z_2)^2 + 4Z_1 Z_2}{(Z_1 + Z_2)^2} \quad (55)$$

$$= \frac{Z_1^2 - 2Z_1 Z_2 + Z_2^2 + 4Z_1 Z_2}{(Z_1 + Z_2)^2} \quad (56)$$

$$= \frac{Z_1^2 + 2Z_1 Z_2 + Z_2^2}{(Z_1 + Z_2)^2} \quad (57)$$

$$= \frac{(Z_1 + Z_2)^2}{(Z_1 + Z_2)^2} \quad (58)$$

$$= 1 \quad \checkmark \quad (59)$$

Special Case: Normal Incidence ($j_1 = j_2 = 0$)

When $j_1 = j_2 = 0$, $\cos j_1 = \cos j_2 = 1$:

From Snell's law at normal incidence, the angles are both zero, and:

$$Z_1 = I_1 = \rho_1 \beta_1, \quad Z_2 = I_2 = \rho_2 \beta_2 \quad (60)$$

$$R_{12} = \frac{I_1 - I_2}{I_1 + I_2} \quad (61)$$

$$T_{12} = \frac{2I_1}{I_1 + I_2} \quad (62)$$

$$\frac{J_R}{J_I} = R_{12}^2 = \left(\frac{I_1 - I_2}{I_1 + I_2} \right)^2 \quad (63)$$

$$\frac{J_T}{J_I} = T_{12}^2 \frac{I_2}{I_1} = \frac{4I_1 I_2}{(I_1 + I_2)^2} \quad (64)$$

8. Summary

Energy density per wavelength: $E = \frac{\rho \omega^2 A^2}{2}$

Energy flux along ray: $J = \frac{\rho \omega^2 A^2}{2} v dS$

Energy flux perpendicular to interface: $J = \frac{\rho \omega^2 A^2}{2} v \cos j dx$

Energy Partition for SH Waves (assuming $I = \rho \beta$):

$$\boxed{\frac{J_R}{J_I} = R_{12}^2} \quad (65)$$

$$\boxed{\frac{J_T}{J_I} = T_{12}^2 \frac{I_2 \cos j_2}{I_1 \cos j_1}} \quad (66)$$

Energy Conservation:

$$\boxed{R_{12}^2 + T_{12}^2 \frac{I_2 \cos j_2}{I_1 \cos j_1} = 1} \quad (67)$$

This completes the energy description of SH-wave propagation across an interface, complementing the kinematic (Snell's Law) and dynamic (amplitude) analyses.

9. Key Points and Summary:

The energy partition formulas show that:

- The amplitude coefficients (R_{12} , T_{12}) are NOT equal to energy partition coefficients. They are **amplitude ratio**, not **energy ratio**.
- Energy partition depends on both the amplitude ratios (R_{12} , T_{12}) and the impedance contrast between media.
- The factor $\frac{I_2 \cos j_2}{I_1 \cos j_1}$ accounts for different energy flux densities in the two media
- At normal incidence, the energy transmission coefficient simplifies to $T_{12}^2 \frac{I_2}{I_1}$
- The reflected energy fraction is simply the square of the reflection coefficient
- The transmitted energy fraction depends on both the transmission coefficient squared AND the impedance ratio
- Energy is perfectly conserved at the interface
- The $\cos j$ factors account for the inclination of the ray path relative to the interface
- For equal impedances ($I_1 = I_2$), all energy is transmitted ($R_{12} = 0$, $T_{12} = 1$).

Part 3 – SH-Waves Post-Critical Reflection

1. Introduction to Post-Critical Reflection

When an SH wave is incident at an angle greater than the critical angle, the transmitted wave becomes evanescent (exponentially decaying with depth). This phenomenon is called **post-critical reflection** or **total internal reflection**.

Critical Angle

The critical angle j_c occurs when the refracted ray would travel parallel to the interface:

We have two elastic solids with properties:

- **Medium 1** ($z < 0$): Shear modulus μ_1 , S-wave velocity β_1
- **Medium 2** ($z > 0$): Shear modulus μ_2 , S-wave velocity β_2

From Snell's law:

$$\frac{\sin j_1}{\beta_1} = \frac{\sin j_2}{\beta_2} \quad (68)$$

At the critical angle, $j_2 = 90$, so $\sin j_2 = 1$:

$$\sin j_c = \frac{\beta_1}{\beta_2} \quad (\text{assuming } \beta_1 < \beta_2) \quad (69)$$

This requires $\beta_1 < \beta_2$ (wave traveling from slower to faster medium).

2. Post-Critical Condition: $j_1 > j_c$

When $j_1 > j_c$:

$$\sin j_2 = \frac{\beta_2}{\beta_1} \sin j_1 > 1 \quad (70)$$

Since $\sin j_2 > 1$, there is no real angle j_2 that satisfies this equation. Instead, we must consider a complex angle.

Complex Wavenumber in Medium 2

The vertical wavenumber in medium 2 becomes imaginary. Recall:

$$k_{z\beta_2}^2 = k_2^2 - k_x^2 = \left(\frac{\omega}{\beta_2}\right)^2 - k_x^2 \quad (71)$$

From continuity of the horizontal wavenumber:

$$k_x = \frac{\omega}{\beta_1} \sin j_1 \quad (72)$$

Therefore:

$$k_{z\beta_2}^2 = \left(\frac{\omega}{\beta_2}\right)^2 - \left(\frac{\omega}{\beta_1}\right)^2 \sin^2 j_1 = \frac{\omega^2}{\beta_2^2} \left(1 - \frac{\beta_2^2}{\beta_1^2} \sin^2 j_1\right) \quad (73)$$

When $j_1 > j_c$, we have $\frac{\beta_2}{\beta_1} \sin j_1 > 1$, making:

$$k_{z\beta_2}^2 < 0 \quad (74)$$

This means $k_{z\beta_2}$ is purely imaginary:

$$k_{z\beta_2} = -i\gamma, \quad \text{where } \gamma > 0 \quad (75)$$

The negative sign ensures the wave decays exponentially in medium 2.

Calculating γ

$$k_{z\beta_2}^2 = \frac{\omega^2}{\beta_2^2} \left(1 - \frac{\beta_2^2}{\beta_1^2} \sin^2 j_1 \right) \quad (76)$$

$$= -\frac{\omega^2}{\beta_2^2} \left(\frac{\beta_2^2}{\beta_1^2} \sin^2 j_1 - 1 \right) \quad (77)$$

$$= -\gamma^2 \quad (78)$$

Therefore:

$$\gamma = \frac{\omega}{\beta_2} \sqrt{\frac{\beta_2^2}{\beta_1^2} \sin^2 j_1 - 1} \quad (79)$$

Using $k_x = \frac{\omega}{\beta_1} \sin j_1$ and $k_{z\beta_1} = \frac{\omega}{\beta_1} \cos j_1$:

$$\gamma = \frac{\omega}{\beta_2} \sqrt{\frac{\beta_2^2 k_x^2}{\omega^2} - 1} = \sqrt{k_x^2 - \frac{\omega^2}{\beta_2^2}} \quad (80)$$

This can also be written as:

$$\gamma = k_x \sqrt{\sin^2 j_1 \left(\frac{\beta_2}{\beta_1} \right)^2 - 1} \quad (81)$$

3. Reflection Coefficient in Post-Critical Case

General Expression

The reflection coefficient from our previous derivation:

$$R_{12} = \frac{\mu_1 k_{z\beta_1} - \mu_2 k_{z\beta_2}}{\mu_1 k_{z\beta_1} + \mu_2 k_{z\beta_2}} \quad (82)$$

Substituting Complex Wavenumber

In the post-critical case, $k_{z\beta_2} = -i\gamma$:

$$R_{12} = \frac{\mu_1 k_{z\beta_1} - \mu_2(-i\gamma)}{\mu_1 k_{z\beta_1} + \mu_2(-i\gamma)} = \frac{\mu_1 k_{z\beta_1} + i\mu_2\gamma}{\mu_1 k_{z\beta_1} - i\mu_2\gamma} \quad (83)$$

Converting to Complex Exponential Form

This can be written as:

$$\boxed{R_{12} = \frac{\mu_1 k_{z\beta_1} + i\mu_2\gamma}{\mu_1 k_{z\beta_1} - i\mu_2\gamma} = e^{i2\epsilon}} \quad (84)$$

where ϵ is the phase shift.

Finding the Phase Shift ϵ

To find ϵ , we use the complex number representation. Let:

$$a = \mu_1 k_{z\beta_1}, \quad b = \mu_2 \gamma \quad (85)$$

Then:

$$R_{12} = \frac{a + ib}{a - ib} \quad (86)$$

To convert this to exponential form, multiply numerator and denominator by the complex conjugate of the denominator:

$$R_{12} = \frac{a + ib}{a - ib} \cdot \frac{a + ib}{a + ib} \quad (87)$$

$$= \frac{(a + ib)^2}{a^2 + b^2} \quad (88)$$

$$= \frac{a^2 - b^2 + 2iab}{a^2 + b^2} \quad (89)$$

The magnitude is:

$$|R_{12}| = \sqrt{\frac{(a^2 - b^2)^2 + (2ab)^2}{(a^2 + b^2)^2}} \quad (90)$$

$$= \sqrt{\frac{a^4 - 2a^2b^2 + b^4 + 4a^2b^2}{(a^2 + b^2)^2}} \quad (91)$$

$$= \sqrt{\frac{a^4 + 2a^2b^2 + b^4}{(a^2 + b^2)^2}} \quad (92)$$

$$= \sqrt{\frac{(a^2 + b^2)^2}{(a^2 + b^2)^2}} = 1 \quad (93)$$

This confirms **total reflection** ($|R_{12}| = 1$).

Alternative Approach for Phase

Using the identity:

$$\frac{a + ib}{a - ib} = \frac{e^{i \arctan(b/a)}}{e^{-i \arctan(b/a)}} = e^{i2 \arctan(b/a)} \quad (94)$$

Therefore:

$$2\epsilon = 2 \arctan \left(\frac{b}{a} \right) = 2 \arctan \left(\frac{\mu_2 \gamma}{\mu_1 k_{z\beta_1}} \right) \quad (95)$$

$$\boxed{\epsilon = \arctan \left(\frac{\mu_2 \gamma}{\mu_1 k_{z\beta_1}} \right) = \tan^{-1} \left(\frac{\mu_2 \gamma}{\mu_1 k_{z\beta_1}} \right)} \quad (96)$$

$$\epsilon = \tan^{-1} \left(\frac{\mu_2 \gamma}{\mu_1 k_{z\beta_1}} \right) = \tan^{-1} \left(\frac{\mu_2 \sqrt{\frac{\sin^2 j_1}{\beta_1^2} - \frac{1}{\beta_2^2}}}{\mu_1 \frac{\cos j_1}{\beta_1}} \right)$$

4. Physical Interpretation

Magnitude of Reflection Coefficient

$$|R_{12}| = \left| \frac{\mu_1 k_{z\beta_1} + i\mu_2 \gamma}{\mu_1 k_{z\beta_1} - i\mu_2 \gamma} \right| = 1 \quad (97)$$

This means **100% of the energy is reflected** - there is no energy transmission into medium 2. This is why it's called "**Total Internal Reflection (TIR)**."

Phase Shift

The reflected wave experiences a phase shift of 2ϵ relative to what would be expected from simple reflection. This phase shift depends on:

- The impedance contrast (μ_1 vs μ_2)
- The angle of incidence (through $k_{z\beta_1}$ and γ)
- How far beyond critical the angle is (through γ)

5. Wave Field in the Medium

Wave Field in Medium 1

The total displacement in medium 1 is:

$$u_{y1} = \underbrace{A_1 e^{i(\omega t - k_x x - k_{z\beta_1} z)}}_{\text{incident}} + \underbrace{A' e^{i(\omega t - k_x x + k_{z\beta_1} z)}}_{\text{reflected}} \quad (98)$$

Since $A' = R_{12} A_1 = e^{i2\epsilon} A_1$:

$$u_{y1} = A_1 e^{i(\omega t - k_x x - k_{z\beta_1} z)} + e^{i2\epsilon} A_1 e^{i(\omega t - k_x x + k_{z\beta_1} z)} \quad (99)$$

$$= A_1 e^{i(\omega t - k_x x)} \left[e^{-ik_{z\beta_1} z} + e^{i2\epsilon} e^{ik_{z\beta_1} z} \right] \quad (100)$$

$$= A_1 e^{i(\omega t - k_x x)} \left[e^{-ik_{z\beta_1} z} + e^{i2\epsilon} e^{ik_{z\beta_1} z} \right] \quad (101)$$

Factoring out $e^{i\epsilon}$:

$$u_{y1} = A_1 e^{i\epsilon} e^{i(\omega t - k_x x)} \left[e^{-i(k_{z\beta_1} z + \epsilon)} + e^{i(k_{z\beta_1} z + \epsilon)} \right] \quad (102)$$

Using $e^{i\theta} + e^{-i\theta} = 2 \cos \theta$:

$$\boxed{u_{y1} = 2A_1 e^{i\epsilon} \cos(k_{z\beta_1} z + \epsilon) e^{i(\omega t - k_x x)}} \quad (103)$$

Alternatively, from:

$$u_{y1} = A_1 e^{i(\omega t - k_x x - k_{z\beta_1} z)} + e^{i2\epsilon} A_1 e^{i(\omega t - k_x x + k_{z\beta_1} z)}$$

Factor the common terms:

$$u_{y1} = A_1 e^{i(\omega t - k_x x)} \left[e^{-ik_{z\beta_1} z} + e^{i2\epsilon} e^{ik_{z\beta_1} z} \right]$$

This represents the interference between incident and reflected waves. The term:

$$e^{-ik_{z\beta_1} z} + e^{i2\epsilon} e^{ik_{z\beta_1} z}$$

describes a standing wave pattern in the z-direction with a phase shift 2ϵ .

$$\boxed{u_{y1} = (1 + e^{i2\epsilon}) A_1 e^{i(\omega t - k_x x - k_{z\beta_1} z)}} \quad (104)$$

This represents a standing wave pattern in the z-direction combined with propagation in the x-direction.

Wave Field in Medium 2 (Evanescent Wave)

The transmitted wave in medium 2 has the form:

$$u_{y2} = A_2 e^{i(\omega t - k_x x - k_{z\beta_2} z)} \quad (105)$$

Substituting $k_{z\beta_2} = -i\gamma$:

$$u_{y2} = A_2 e^{i(\omega t - k_x x - (-i\gamma)z)} \quad (106)$$

$$u_{y2} = A_2 e^{i(\omega t - k_x x)} e^{(i \cdot i \gamma z)} \quad (107)$$

$$\boxed{u_{y2} = A_2 e^{-\gamma z} e^{i(\omega t - k_x x)}} \quad (108)$$

Characteristics of Evanescent Wave

- **Exponential decay:** Amplitude decreases exponentially with depth as $e^{-\gamma z}$
- **Propagates horizontally:** The wave propagates along the interface (x-direction) with wavenumber k_x
- **No vertical propagation:** No energy is transmitted downward into medium 2
- **Decay length:** The characteristic decay length is $1/\gamma$

6. Energy Considerations

Energy Flux in Medium 2

The energy flux in the z-direction in medium 2 is:

$$J_{z2} = \text{Re} [\sigma_{yz2} \dot{u}_{y2}^*] \quad (109)$$

For the evanescent wave:

$$\dot{u}_{y2} = i\omega A_2 e^{-\gamma z} e^{i(\omega t - k_x x)} \quad (110)$$

$$\sigma_{yz2} = \mu_2 \frac{\partial u_{y2}}{\partial z} = -\mu_2 \gamma A_2 e^{-\gamma z} e^{i(\omega t - k_x x)} \quad (111)$$

Note that σ_{yz2} is real (no imaginary time factor), so:

$$J_{z2} \propto \text{Re}[\text{real} \times \text{imaginary}] = 0 \quad (112)$$

No energy is transmitted in the z-direction, confirming total reflection.

Verification: $|R_{12}| = 1$

Using the energy partition formula:

$$E_R = \frac{J_R}{J_I} = |R_{12}|^2 = 1 \quad (113)$$

This means 100% of incident energy is reflected.

7. Summary

Post-Critical Reflection Conditions:

$$j_1 > j_c \quad \text{where} \quad \sin j_c = \frac{\beta_1}{\beta_2} \quad (114)$$

Complex Wavenumber:

$$k_{z\beta_2} = -i\gamma, \quad \gamma = \sqrt{k_x^2 - \frac{\omega^2}{\beta_2^2}} > 0 \quad (115)$$

Reflection Coefficient:

$$R_{12} = \frac{\mu_1 k_{z\beta_1} + i\mu_2 \gamma}{\mu_1 k_{z\beta_1} - i\mu_2 \gamma} = e^{i2\epsilon} \quad (116)$$

Phase Shift:

$$\epsilon = \tan^{-1} \left(\frac{\mu_2 \gamma}{\mu_1 k_{z\beta_1}} \right) \quad (117)$$

Key Properties:

- $|R_{12}| = 1$ (total reflection)
- Phase shift 2ϵ in reflected wave
- Evanescent wave in medium 2: $u_{y2} \propto e^{-\gamma z} e^{i(\omega t - k_x x)}$
- No energy transmission into medium 2
- Standing wave pattern in medium 1 along z-direction
- Wave propagates along interface in both media
- For post-critical incidence, the SH-wave reflection coefficient becomes $R_{12} = e^{i2\epsilon}$, representing total reflection with a frequency-dependent phase shift, while the transmitted wave becomes an evanescent wave that decays exponentially away from the interface.

Special Cases

- **At critical angle** ($j_1 = j_c$): $\gamma = 0$, so $\epsilon = 0$, $R_{12} = 1$
- **At grazing incidence** ($j_1 \rightarrow 90^\circ$): $k_{z\beta_1} \rightarrow 0$, so $\epsilon \rightarrow \frac{\pi}{2}$, $R_{12} \rightarrow -1$
- The phase shift varies smoothly from 0 to π as incidence angle increases from critical to grazing

This completes the mathematical description of post-critical SH-wave reflection, showing how the reflection coefficient becomes a pure phase factor and all energy is totally reflected.